Merged mathematical equations of atmosphere and pollution

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Abstract

This document describes the outline of this project's plan.

Contents

1 Equations of the atmosphere (3D, z - p variable change, without terrain-influenced coordinate-change)

- Comments with a bullet sign mean questions / ideas / thoughts to be discussed.

The 3-dimensional equations of the atmosphere. Two horizontal dimensions, one vertical dimension (using the coordinate change p - z), mass continuity, equation of state and amount of water. Equations from the Temam-Ziane article (p34). Variables: (x, y, p).

$$\frac{\partial \mathbf{v}}{\partial t} + \nabla_{\mathbf{v}} \mathbf{v} + \omega \frac{\partial \mathbf{v}}{\partial p} + 2\Omega \sin\theta \mathbf{k} \times \mathbf{v} + \nabla \Phi - L_{\mathbf{v}} \mathbf{v} = F_{\mathbf{v}}$$

$$\frac{\partial \Phi}{\partial p} + \frac{RT}{p} = 0$$

$$\nabla \cdot \mathbf{v} + \frac{\partial \omega}{\partial p} = 0$$

$$\frac{\partial T}{\partial t} + \nabla_{\mathbf{v}} T + \omega \frac{\partial T}{\partial p} - \frac{R\bar{T}}{c_p p} \omega - L_T T = F_T$$

$$\frac{\partial q}{\partial t} + \nabla_{\mathbf{v}} q + \omega \frac{\partial q}{\partial p} - L_q q = F_q$$

$$p = R\rho T$$

With Ω being the angular velocity, L_{arb} a Laplace operator, Φ geopotential.

- Dry air is a perfect gas, but humid air is not. For dry air the gases in the atmosphere have a simple equation of state known as the ideal gas law.

- For moist air, the gas constant changes because water vapor is less dense than dry air. To simplify things, a virtual temperature, T_v can be defined to include the effects of water vapor on density: $T_v = T(1 + 0.61w)$, where w is the water vapor mixing ratio (g water vapor / kg dry air) and all temperatures are in K. Moist air of temperature T behaves as dry air of temperature T_v.

2 Chemical kinetics - equations describing pollution (3D, without terrain-influenced coordinate-change)

Species continuity equation, equations of chemical kinetics. See the green Temam book chapter 11 and the onlinelibrary article.

$$\frac{\partial C_j}{\partial t} + \nabla \cdot J_j = r_j$$

where C_j is the local molar concentration of species j and the components of J_j represent the local molar flux and r_j is a volume based formation rate. By nature it describes a connection between time variations in the species concentration in a fixed point, the local motion of the species, and the rate the species is formed by chemical reactions.

Rectangular coordinates:

$$\frac{\partial C_j}{\partial t} + \frac{\partial J_{jx}}{\partial x} + \frac{\partial J_{jy}}{\partial y} + \frac{\partial J_{jz}}{\partial z} = r_j$$

See more: Species Continuity Equation Link

In a slightly different, more expanded form the same thing, $1 \leq i \leq N$:

$$\frac{\partial}{\partial t}(\rho Y_i) + \nabla \cdot (\rho Y_i u_i) = \omega_i,$$

where Y_i is the mass fraction of the species i and ω_i is a term of sink-source nature.

3 ONE. Creating a synchronized system of an atmosphere containing pollution (3D, (x, y, p), without terrain-influenced coordinate-change)

The first two chapters describe two independent / autonomous 3D systems in different situations, with independent solutions. The first task is to merge the two models coming from different places so that they make sense in a jointed sense (still in 3D), and have common variables, common solution.

Possible / common approach for merging (Temam book p174 and p176):

i) on one hand we write the fluid equations of the mixture considered as a single fluid with denisty ρ_0, pressure p, temperature T, velocity $\mathbf{v}$,

ii) on the other hand we consider the equations describing the evolution of each mass fraction (or species concentration) Y_i or (C_i).

The unknown functions are velocity $\mathbf{v}$ of the mixture, its pressure p, the temperature T, and the mass densities $Y_i = \frac{\rho_i(t,x)}{\rho}$ (Eulerian description), $1 \leq i \leq N$. Equations: N-S, temperature equation of the mixture, species continuity equation.

Changes: $\rho = \rho_0$ homogeneous fluid, instead of u_i in the continuity equation we use the notation $\mathbf{v}_i$. Since the equations of the atmosphere in this case are in (x, y, p) coordinates, we use the J_{zp} Jacobian in the species continuity equation to sync.

$$\frac{\partial \mathbf{v}}{\partial t} + \nabla_{\mathbf{v}}\mathbf{v} + \omega\frac{\partial \mathbf{v}}{\partial p} + 2\Omega\sin\theta\mathbf{k}\times\mathbf{v} + \nabla\Phi - L_{\mathbf{v}}\mathbf{v} = F_{\mathbf{v}}$$

$$\frac{\partial \Phi}{\partial p} + \frac{RT}{p} = 0$$

$$\nabla\cdot\mathbf{v} + \frac{\partial \omega}{\partial p} = 0$$

$$\frac{\partial T}{\partial t} + \nabla_{\mathbf{v}}T + \omega\frac{\partial T}{\partial p} - \frac{R\bar{T}}{c_p p}\omega - L_T T = F_T$$

$$\frac{\partial q}{\partial t} + \nabla_{\mathbf{v}}q + \omega\frac{\partial q}{\partial p} - L_q q = F_q$$

$$p = R\rho_0 T$$

$$\frac{\partial}{\partial t}(\rho_0 Y_i J_{zp}) + \nabla\cdot(\rho_0 J_{zp} Y_i \mathbf{v}_i) = \omega_i J_{zp},\ 1 \le i \le N$$

$$\rho_0 = \sum_{i=1}^{N}\rho_i, (\mathbf{v},\omega) = \sum_{i=1}^{N}\frac{\rho_i}{\rho_0}\mathbf{v_i}J_{zp}$$

Remark. *In the above equations ω is a notation for the speed of the wind in the common atmosphere equations, while ω_i stands for the production rate of species in the continuity equation.*

- In the Temam book the null divergence condition is assumed but in this case we are talking about air, which is compressible

4 Terrain influenced coordinates

A more general version of the species continuity equation describing chemical kinetics is a version including terrain-influenced coordinates. The terrain-influenced coordinate η is defined as the following (see the Linus Magnusson article):

$$\eta(x,y,z) = s\frac{z - z_G(x,y)}{s - z_G(x,y)},$$

where s is the top height of the model and z_G is the height of the ground.

Calculation of the J_η Jacobian matrix of changing from (x,y,z) to (x,y,η):

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{\partial z_G}{\partial x}(\frac{\eta - s}{s - z_G}) & \frac{\partial z_G}{\partial y}(\frac{\eta - s}{s - z_G}) & \frac{s}{s - z_G} \end{pmatrix}$$

Using this J_η Jacobian matrix of the coordinate change on the chemical kinetics equation and separate horizontal and vertical velocity it gives

$$\frac{\partial(\rho_0 Y_i J_\eta)}{\partial t} + \nabla_\eta \cdot (\rho_0 Y_i J_\eta \mathbf{v}_{horiz}) + \frac{\partial \rho_0 Y_i J_\eta \mathbf{v}_{vert}}{\partial \eta} = J_\eta \omega_i.$$

- The role of m^2 in the ScienceDirect paper, ∇_η is not linear or why do the m^2 items don't cancel each other.
- The correctness of the new equation with J_η.

This is basically the same where the ScienceDirect equation is.

Like this it can not be matched with the original system, since that uses pressure and this uses terrain-influenced transformed vertical coordinates.

5 TWO. A way of matching chemical kinetics equation in terrain influenced coordinates and equations of the atmosphere. Variables: (x, y, η)

One thing we can do is to consider the set of the equations of the atmosphere in classical cartesian coordinates (instead of taking p for vertical coordinate instead of z), use the terrain influenced coordinate change on the equations of the atmosphere and couple them with the kinetics equation from the previous section. (This might also be convenient because the "Mathematical justification..." paper is also does not use the pressure instead of z approach.)

Equations describing the motion of the atmosphere as one single mixture:

$$\frac{\partial \mathbf{v} J_\eta}{\partial t} + \nabla_\mathbf{v} \mathbf{v} J_\eta + w J_\eta \frac{\partial \mathbf{v} J_\eta}{\partial z} + \frac{1}{\rho_0} \nabla p J_\eta + 2\Omega \sin\theta \mathbf{k} \times \mathbf{v} J_\eta - \mu_\mathbf{v} \Delta \mathbf{v} J_\eta - \nu_\mathbf{v} \frac{\partial^2 \mathbf{v} J_\eta}{\partial z^2} = 0$$

$$\frac{\partial p J_\eta}{\partial z} = -\rho_0 g$$

$$\frac{\partial \rho_0}{\partial t} + \rho_0 (\nabla \mathbf{v} + \frac{\partial w J_\eta}{\partial z}) + \mathbf{v} \nabla \rho_0 + w \frac{\partial \rho_0}{\partial z} = 0$$

$$\frac{\partial T J_\eta}{\partial t} + \nabla_\mathbf{v} T J_\eta + w J_\eta \frac{\partial T J_\eta}{\partial z} - \mu_T \Delta T J_\eta - \nu_T \frac{\partial^2 T J_\eta}{\partial z^2} - \frac{R T J_\eta}{p J_\eta} \frac{\mathrm{d} p J_\eta}{\mathrm{d} t} = Q_T$$

$$\frac{\partial q J_\eta}{\partial t} + \nabla_\mathbf{v} q J_\eta + w J_\eta \frac{\partial q J_\eta}{\partial z} - \mu_q \Delta q J_\eta - \nu_q \frac{\partial^2 q J_\eta}{\partial z^2} = 0$$

$$p J_\eta = R \rho_0 T J_\eta$$

Chemical kinetics equation describing pollution as individual motions of N species:

$$\frac{\partial(\rho_0 Y_i J_\eta)}{\partial t} + \nabla_\eta \cdot (\rho_0 Y_i J_\eta \mathbf{v}) + \frac{\partial \rho_0 Y_i J_\eta w}{\partial \eta} = J_\eta \omega_i,$$

$$\rho_0 = \sum_{i=1}^{N} \rho_i, p = \sum_{i=1}^{N} p_i, \mathbf{v} J_\eta = \sum_{i=1}^{N} \frac{\rho_i}{\rho_0} \mathbf{v_i} J_\eta, w J_\eta = \sum_{i=1}^{N} \frac{\rho_i}{\rho_0} w J_\eta$$

Remark. *The equation above describing ρ is kept for now as it is in the original set, however ρ is assumed to be constant ρ_0, so most part of the equation disappears.*

6 THREE. Sync in cartesian coordinates. Variables: (x, y, z)

Take the original chemical kinetics equation in cartesian coordinates and couple it with the equations of the atmosphere without terrain-influenced coordinate change and without using the exponentially (monotonously) decreasing p-z relationship. Here we use the notation $(\mathbf{v}, w)_i$ for the species velocities instead of u_i.

$$\frac{\partial \mathbf{v}}{\partial t} + \nabla_{\mathbf{v}} \mathbf{v} + w \frac{\partial \mathbf{v}}{\partial z} + \frac{1}{\rho_0} \nabla p + 2\Omega \sin\theta \mathbf{k} \times \mathbf{v} - \mu_{\mathbf{v}} \Delta \mathbf{v} - \nu_{\mathbf{v}} \frac{\partial^2 \mathbf{v}}{\partial z^2} = 0$$

$$\frac{\partial p}{\partial z} = -\rho_0 g$$

$$\frac{\partial \rho_0}{\partial t} + \rho_0 (\nabla \mathbf{v} + \frac{\partial w}{\partial z}) + \mathbf{v} \nabla \rho_0 + w \frac{\partial \rho_0}{\partial z} = 0$$

$$\frac{\partial T}{\partial t} + \nabla_{\mathbf{v}} T + w \frac{\partial T}{\partial z} - \mu_T \Delta T - \nu_T \frac{\partial^2 T}{\partial z^2} - \frac{RT}{p} \frac{\mathrm{d}p}{\mathrm{d}t} = Q_T$$

$$\frac{\partial q}{\partial t} + \nabla_{\mathbf{v}} q + w \frac{\partial q}{\partial z} - \mu_q \Delta q - \nu_q \frac{\partial^2 q}{\partial z^2} = 0$$

$$p = R \rho_0 T$$

$$\frac{\partial}{\partial t}(\rho Y_i) + \nabla \cdot (\rho Y_i (\mathbf{v}, w)_i) = \omega_i, \ 1 \le i \le N$$

$$\rho_0 = \sum_{i=1}^{N} \rho_i, p = \sum_{i=1}^{N} p_i, (\mathbf{v}, w) = \sum_{i=1}^{N} \frac{\rho_i}{\rho_0} (\mathbf{v}, w)_i$$

7 Choosing and refining the model

We have seen different approaches for dealing with the vertical coordinate, such as:

i) leaving it as classical cartesian coordinate

ii) changing to pressure as vertical coordinate because of well-behaving monotonous relationship between height and pressure,

iii) introducing the terrain-influenced vertical coordinate.

But eventually we will do dimension reduction because of the quasi-horizontal nature of the atmosphere, so it is very useful and convenient to consider the one in cartesian coordinates (see THREE).

For refining the model we are considering the approach described in the ScienceDirect article regarding the chemical kinetics equation.

In the ScienceDirect article we start from the continuity equation in the following form:

$$\frac{\partial(\varphi_i J_s)}{\partial t} + m^2 \nabla_s \cdot (\frac{\varphi_i J_s V_s}{m^2}) + \frac{\partial(\varphi_i J_s s)}{\partial s} = J_s Q_\varphi$$

8 Dimension reduction for the jointed system

The 3D model is too complex to work with, on the other hand the phenomena can be seen as quasi-horizontal, which is why we perform dimension reduction for the merged system, the goal is to have a synchronized 2D system describing pollution and atmosphere.

9 Solutions for the jointed two-dimensional system

In this chapter we investigate solutions for the new, two-dimensional merged system.

10 Numerical simulations for the jointed two-dimensional system

Finally we perform numerical simulations.

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